

Faculty Research Grant & Publication Tracker

Complete Requirements Table — Problem Statement #09

Functional Requirements

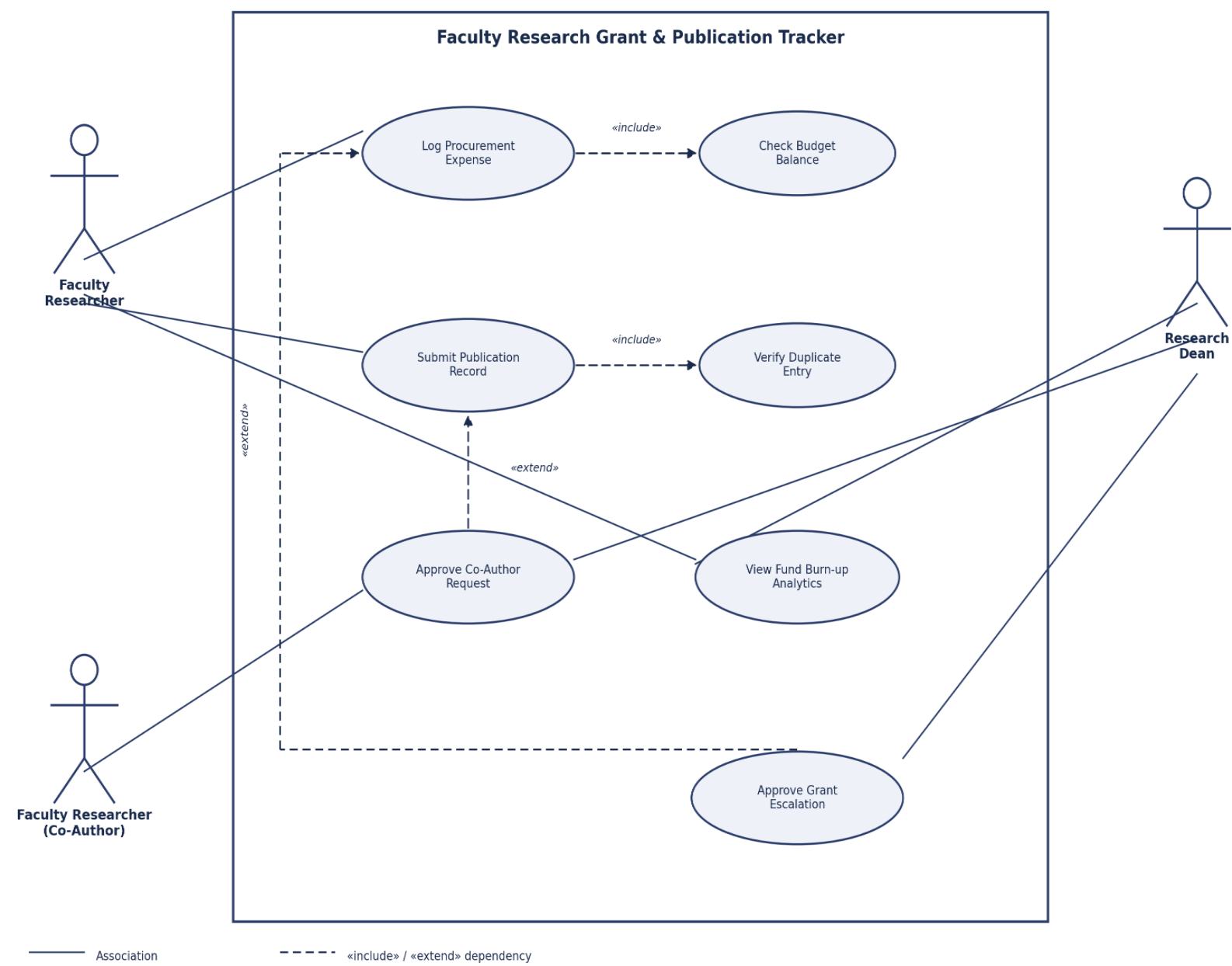
ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall track grant fund allocations, allow faculty to log equipment/consumable procurement expenses against a grant, and compute the remaining budgetary balance in real time.	High	Pass: Balance updates accurately after each expense and the system blocks any expense that would exceed the grant cap unless an admin override is applied. Fail: Overdraft is allowed without an admin override, or the displayed balance does not match the sum of allocations minus expenses.	Prevents grant overdrafts and gives faculty real-time visibility into spendable funds, which is the core financial control the research deanery needs.
FR-002	The system shall allow faculty to submit publication records including title, authors, venue, DOI, indexing status (Scopus/WoS/UGC-CARE), and citation count, and shall check each submission for duplicate DOI/title entries before acceptance.	High	Pass: A duplicate DOI or high-similarity title is detected and the submission is blocked with a clear conflict message; unique submissions are stored successfully. Fail: A duplicate record is silently created, or a valid unique submission is rejected.	Accurate, de-duplicated publication data is essential for correct citation/indexing metrics reported to the deanery and external ranking bodies.
FR-003	For publications with more than one author, the system shall route a co-author approval request to each declared co-author and shall only mark the publication as “verified” once every co-author has approved the authorship order and contribution share.	Medium	Pass: Publication status changes to “verified” only after 100% of co-authors approve; a single rejection reverts the record to draft. Fail: A publication is marked verified while a co-author approval is still pending or rejected.	Protects academic integrity and ensures all contributing authors have consented to how credit is recorded before it counts toward departmental metrics.
FR-004	The system shall generate fund burn-up analytics, visually plotting cumulative grant expenditure against sanctioned budget and time-based milestones, filterable by grant, category, and date range.	Medium	Pass: The burn-up chart renders correct cumulative totals matching the underlying ledger for any selected filter combination. Fail: Chart totals diverge from the ledger, or filters produce no data / incorrect date ranges.	Gives the Research Dean an at-a-glance view of spending velocity across grants, supporting proactive budget management ahead of grant closure deadlines.
FR-005	The system shall automatically notify the Research Dean whenever a grant's cumulative fund utilization crosses the 75%, 90%, or 100% threshold, and shall require Dean approval before further expenses can be logged past the breached threshold.	High	Pass: A notification is generated within the same transaction that crosses a threshold, and further expense logging is blocked until the Dean approves or rejects escalation. Fail: Expenses continue to be logged past a threshold with no notification or approval gate.	Provides an early-warning and control mechanism so grants are not exhausted or overdrawn without deanery visibility and sign-off.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Security & Auditability	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals (expense logging, escalation decisions) and journal/publication status modifications, recording actor, timestamp, and before/after state.	High	Pass: Benchmarking and penetration tests confirm every financial/status-change event produces a cryptographically verifiable ledger entry that cannot be edited or deleted, under simulated peak load. Fail: Any ledger entry can be silently altered, or an approval/status change occurs without a corresponding entry.	Sponsored research funds and academic records require a defensible, tamper-proof trail for internal audits and external funding-agency compliance reviews.

NFR-002	Availability & Scalability	The system shall be available with at least 99.5% uptime during active grant-submission and publication-reporting windows, and shall support at least 500 concurrent authenticated users without degradation of response time beyond 2 seconds for standard read operations.	Medium	Pass: Load testing at 500 concurrent users maintains sub-2-second response for standard reads and uptime monitoring shows ≥99.5% availability over a rolling 30-day window. Fail: Response times exceed target under load, or uptime falls below 99.5% during a reporting window.	The portal must remain responsive during peak periods such as grant-renewal deadlines and annual publication reporting cycles, when faculty and deanery usage spikes simultaneously.
---------	----------------------------	--	--------	--	--

UML USE CASE DIAGRAM:



Brief Description

An optional extension triggered only when a logged expense causes cumulative grant utilization to cross a configured threshold (75%, 90%, or 100%). The Research Dean must review and approve continued spending beyond the threshold before further expenses can be logged against the grant.

Preconditions

- A grant's cumulative utilization has just crossed a configured escalation threshold as a result of an expense entry.
- The extension point “Threshold Breach” was reached during Log Procurement Expense.

Postconditions

- The escalation decision (Approved / Rejected / Approved with revised cap) is recorded against the grant.
- If approved, further expense logging against the grant resumes normally up to the next threshold.
- If rejected, the grant is flagged as “spending frozen” pending deanery review.
- An audit-ledger entry captures the escalation decision, decision-maker, and timestamp.

Main Success Scenario

- 1 System (extending Log Procurement Expense at the “Threshold Breach” extension point) detects utilization has crossed a threshold and generates an escalation request.
- 2 System notifies the Research Dean with grant ID, threshold crossed, and remaining balance.
- 3 Research Dean opens the escalation request and reviews recent expense history and burn-up trend.
- 4 Research Dean approves continued spending, optionally adjusting the category-wise cap.
- 5 System records the approval, updates the grant's active threshold, and resumes normal expense logging.

6 System notifies the Faculty Researcher that spending has been re-authorized.

Alternate Flow — Dean Rejects Escalation

- 1 At step 4, Research Dean determines further spending is not justified and selects “Reject.”
- 2 System flags the grant as “spending frozen” and blocks all further Log Procurement Expense transactions.
- 3 System notifies the Faculty Researcher of the freeze and the reason provided by the Research Dean.
- 4 Faculty Researcher may submit a written justification, restarting the escalation review at step 3.

Exception / Error Flow

If the Research Dean does not act within the SLA window, the system auto-freezes the grant as a safety default and logs the SLA breach in the audit ledger for compliance review.

Notes / UML Relationship Detail

Modeled with an «extend» relationship because escalation approval is conditional — it only fires when a threshold is crossed — and it augments the base Log Procurement Expense flow at a well-defined extension point rather than executing on every invocation.